

ANTHROPIC

Appendix: Coding Agents in the Social Sciences

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Data description

The survey was fielded from February 20 to March 24, 2026. We identified respondents in five ways. First, we scraped faculty and student contact information from sociology, political science, management, economics, psychology, education, and public health department websites at R1 and major Canadian research universities. Second, we used OpenAlex, an open source repository of academic research, to identify research-active social scientists based on publication and working paper topics. We restricted to researchers who had published within the last year, in hopes of identifying active, working researchers. We also used academic conference programs for the same purpose. We directly emailed 44,700 academics we identified in these ways. Additionally, we asked directors of graduate studies in sociology, political science, and economics to distribute invitations to their graduate students, and we distributed invitations via the message boards and email lists of scholarly communities such as the Academy of Management and the American Sociological Association.

Throughout these recruitment efforts, we targeted researchers based in the US or Canada actively doing quantitative, empirical research. In an opening screener for the survey, we further restricted participants to those who work with quantitative empirical data, and, among doctoral students, those who have finished at least 2 years of their training and who plan to enter academia. Invitations were sent by email. Respondents were informed that the study concerns academic workflows and that participants would be eligible for a randomized experiment providing access to Claude Max accounts. They were also offered a \$10 gift card for completing the survey. After applying the screener and limiting to respondents who completed the full survey, the analytic sample consists of 1,260 respondents, described by discipline and career stage in Table A1.

Table A1: Sample of respondents to the WFL Baseline Survey, by career stage and discipline

Career stage	N	%	Discipline	N	%
PhD Student	384	31	Economics	209	17
Postdoc	87	7	Political Science	216	17
Assistant Professor	311	25	Sociology	262	21
Associate Professor	235	19	Psychology	172	14
Full Professor	243	19	Management Sciences	202	16
			Public Health	84	7
			Education	45	4
			Communication	36	3
			Other	34	3
Total	1,260	100	Total	1,260	100

Respondents provided their discipline via free text entry. The raw responses contain 318 unique values, many of which are idiosyncratic subfield descriptions. We consolidate these into eight substantive discipline buckets plus a residual "Other" category, using a mix of Claude classification and hand coding. Economics includes its applied subfields (health economics, development economics) and political economy. Political science includes public administration and law. Sociology includes urban studies, geography and social work. Management sciences includes all applied disciplines typically housed in business schools (including finance, accounting, and organizational behavior). Public health includes epidemiology. "Other" is a residual category, including the small number of respondents doing quantitative research in anthropology, archaeology, history and linguistics.

We infer gender from each respondent's self-reported first name using the `gender_guesser` Python library. We collapse "male" and "mostly_male" into a single "typically male name" category, and "female" and "mostly_female" into "typically female name." Unknown or androgynous names are left out of the analysis.

We classify respondents' institutions as top 25 in the US and Canada using the Nature Index 2025 Leading Institutions ranking. 28% of respondents are from the top 25 institutions, which include Harvard University, Stanford University, MIT, National Institutes of Health, University of Michigan, Yale University, University of Pennsylvania, University of Toronto, UCLA, UC Berkeley, Johns Hopkins University, University of Washington, UC San Diego, Columbia University, Cornell University, Northwestern University, University of

Chicago, Washington University in St. Louis, Princeton University, Caltech, UC San Francisco, UT Austin, University of Wisconsin-Madison, Duke University, and University of Minnesota.

The adjusted output comparisons in Figure 5b use OLS regressions of the form:

$$Output_i = \beta \times CodingAgentUser_i + \gamma \times X_i + \epsilon_i$$

where X_i includes fixed effects for career stage (five categories: PhD student, postdoc, assistant professor, associate professor, full professor), discipline bucket (nine categories), and the calendar week the respondent completed the survey. Standard errors are heteroskedasticity-robust. The coefficient beta is the adjusted difference in mean output between coding agent users and non-users, conditional on career stage, discipline, and survey timing. We fit this model with OLS, and then divide by the control group's mean of each outcome to back out percent differences. A Poisson model specification gives similar results.

The raw differences in output among respondents are more difficult to interpret, as they reflect differences between disciplines in practices like posting working papers, timing of conference submissions, and expected number of journal publications per year. These unadjusted differences are shown in Figure A1 and show that agent users post more working papers and submit more grants, but actually submit fewer papers to journals. However, as Figure 5 shows, this lower rate of submitting to journals reflects the distribution of coding agent use across disciplines and career stages.

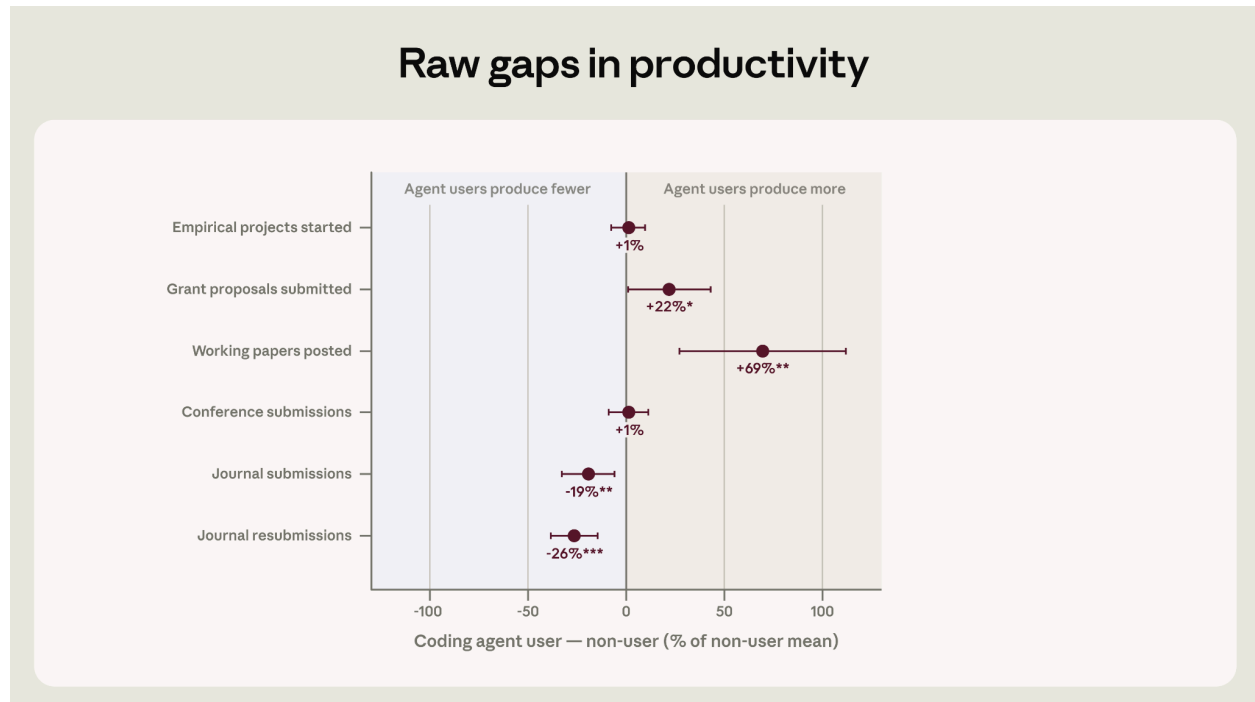


Figure A1: Unadjusted research productivity differences between coding agent users and other researchers. Outcomes are self-reported by respondents. Working papers posted, journal submissions and journal resubmissions are mutually exclusive categories. See Figure 5 for estimates adjusted by discipline, career stage and survey week. Whiskers show 95% confidence intervals calculated using the robust standard errors scaled by the outcome mean. * $p < 0.05$, ** $p < 0.01$.

Pilot sample

How much is selection into the survey biasing results? The main sample we use here suffers from its function as recruitment for an experiment on Claude Code access and research productivity. To assess how important this distortion is, we compare results in this sample to those from a pilot we ran in early February. This pilot was much smaller, explicitly excluded full professors, and only targeted sociologists, political scientists and management scholars. In the main sample, we expanded disciplines and career stage to draw in a broader range of respondents. The pilot survey also had some different questions on it, as we piloted and improved the survey. Crucially, it is missing our detailed coding agent question.

But the pilot data are useful because we did not advertise the subsequent experiment in recruitment for it. Instead we simply described a study on changes in research workflows.

This means we can use the pilot data to assess whether our results in the main sample are driven by selection into the experiment.

Table A2 shows results comparing the pilot sample to the full survey. To make the recruitment logic of the two samples comparable, we narrow the main survey sample to sociologists, political scientists and management scholars and exclude full professors. There is little difference in overall rates of AI use across the two samples, but the full sample uses AI for more research tasks and is more optimistic about its impact. However, the key gaps persist: coding and editing are the most common uses in both samples and drafting prose is rare in both. On the 1 to 10 slider, the full sample is one point more optimistic about field impacts and a half point more optimistic about paper productivity effects. But the gap between paper productivity and field impact beliefs persist. Overall, the full sample is composed of more pro-AI respondents, but the key gradients that we can measure are similar in the pilot sample drawn without AI-related recruitment materials.

Table A2: Comparison of main and pilot samples

	Main sample		Pilot sample		Difference
	Mean	N	Mean	N	
AI adoption					
AI use	84.5%	446	87.3%	102	+2.7pp
Beliefs about AI (1-10 scale)					
Productivity	7.56	446	7.10	102	-0.46*
Field impact	5.68	446	4.75	102	-0.93***
AI use cases (among AI users)					
Code	84.1%	377	76.4%	89	-7.7pp
Edit prose	74.8%	377	68.5%	89	-6.3pp
Method advice	69.2%	377	64.0%	89	-5.2pp
Lit review	66.0%	377	40.4%	89	-25.6pp***
Generate ideas	41.1%	377	36.0%	89	-5.2pp
Draft prose	35.8%	377	27.0%	89	-8.8pp
N use cases (mean)	3.71	377	3.12	89	-0.59**

Notes: The main sample is restricted to include only sociology, political science, and management and exclude full professors, for clearer comparison to the narrow pilot sample. Use case shares are conditional on AI use.